MILITARY SPECIFICATION SHEET

WAVEGUIDES, RIGID, RECTANGULAR (MILLIMETER WAVELENGTH)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguides described herein shall consist of this document and the latest issue of Specification MIL-W-85.

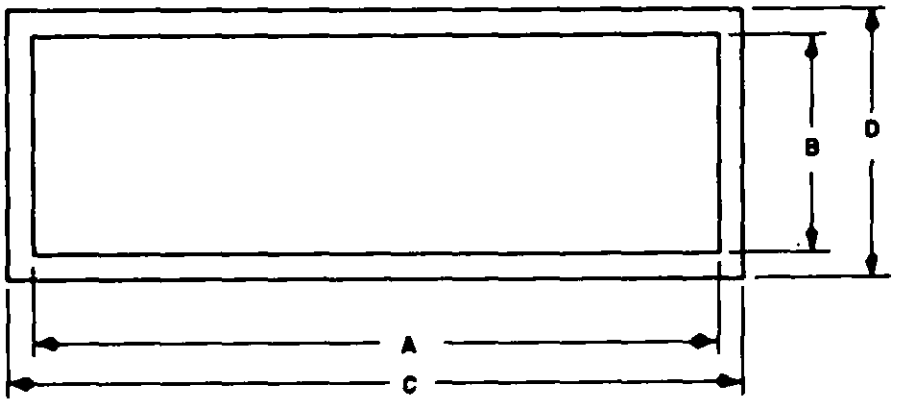


Figure 1-Waveguide, Rigid, Rectangular (Millimeter Wavelength).

REQUIREMENTS:
Design, construction, and physical dimensions: See figure 1 and table I.
Material: See table I.
Part number: See table I.

NOTES:
Engineering information: See table II.
Supersession data and cross index: See table III.
Typical resistivity and conductivity of materials: See table IV.

FSC 5985
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TABLE I. Waveguides, rigid, rectangular (millimeter wavelength).

Part 1/ number M85/3-	Dimensions (inches) 2/						Wall Thickness (Nominal)	Outer corner radius (max.)	Inner corner radius (max.)	ASTM type material
	Inside			Outside						
	Width (A)	Height (B)	Tolerance	Width (C)	Height (D)	Tolerance				
006-XXX 007-XXX 008-XXX 009-XXX	.2800 (7.11)	.1400 (3.56)	±.0015 (.04)	.360 (9.14)	.220 (5.59)	±.002 (.05)	.040 (1.02)	.012 (.30)	.008 (.20)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282 6061
010-XXX 011-XXX 012-XXX 013-XXX	.2240 (5.69)	.1120 (2.84)	±.0010 (.03)	.304 (7.72)	.192 (4.88)	±.002 (.05)	.040 (1.02)	.010 (.25)	.007 (.18)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282 6061
014-XXX 015-XXX 016-XXX	.1880 (4.78)	.0940 (2.39)	±.0010 (.03)	.268 (6.81)	.174 (4.42)	±.002 (.05)	.040 (1.02)	.009 (.23)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
017-XXX 018-XXX 019-XXX	.1480 (3.76)	.0740 (1.88)	±.0010 (.03)	.228 (5.79)	.154 (3.91)	±.002 (.05)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
020-XXX 021-XXX 022-XXX	.1220 (3.10)	.0610 (1.55)	±.0010 (.03)	.202 (5.13)	.141 (3.58)	±.002 (.05)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
023-XXX 024-XXX 025-XXX	.1000 (2.54)	.0500 (1.27)	±.0010 (.03)	.180 (4.57)	.130 (3.30)	±.002 (.05)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
026-XXX 027-XXX 028-XXX	.0800 (2.03)	.0400 (1.02)	±.0005 (.01)	.1600 (4.06)	.1200 (3.09)	±.0015 (.04)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
029-XXX 030-XXX 031-XXX	.0650 (1.65)	.0325 (.83)	±.0005 (.01)	.1450 (3.68)	.1125 (2.86)	±.0015 (.04)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
032-XXX 033-XXX 034-XXX	.0510 (1.30)	.0255 (.65)	±.0005 (.01)	.1310 (3.33)	.1055 (2.68)	±.0015 (.04)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282

See footnotes at end of table.

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TABLE I. Waveguides, rigid, rectangular (millimeter wavelength) - Continued.

Part 1/ number M85/3-	Dimensions (inches) 2/									ASTM type material
	Inside			Outside			Wall Thickness (Nominal)	Outer corner radius (max.)	Inner corner radius (max.)	
	Width (A)	Height (B)	Tolerance	Width (C)	Height (D)	Tolerance				
035-XXX 036-XXX 037-XXX	.0430 (1.09)	.0215 (.55)	±.0005 (.01)	.1230 (3.12)	.1015 (2.58)	±.0015 (.04)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
038-XXX 039-XXX 040-XXX	.0340 (.86)	.0170 (.43)	±.0005 (.01)	.1140 (2.90)	.0970 (2.46)	±.0015 (.04)	.040 (1.02)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
041-XXX 042-XXX 043-XXX	.0800 (2.03)	.0400 (1.02)	±.0005 (.01)	.120 (3.05)	.080 (2.03)	±.0010 (.03)	.020 (.51)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
044-XXX 045-XXX 046-XXX	.0650 (1.65)	.0325 (.83)	±.0005 (.01)	.105 (2.67)	.0725 (1.83)	±.0010 (.03)	.020 (.51)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
047-XXX 048-XXX 049-XXX	.0510 (1.30)	.0255 (.65)	±.0005 (.01)	.091 (2.31)	.0655 (1.65)	±.0010 (.03)	.020 (.51)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
050-XXX 051-XXX 052-XXX	.0430 (1.09)	.0215 (.55)	±.0005 (.01)	.083 (2.11)	.0615 (1.55)	±.0010 (.03)	.020 (.51)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282
053-XXX 054-XXX 055-XXX	.0340 (.86)	.0170 (.43)	±.0005 (.01)	.074 (1.88)	.057 (1.45)	±.0010 (.03)	.020 (.51)	.008 (.20)	.006 (.15)	MIL-S-13282 Grade C OF - DLP Copper alloy 3/ and MIL-S-13282

1/ The complete part number shall be the number shown and a three-digit number indicating length in inches (for example, M85/3-006-096). Part numbers M85/3-001 through -005 have been deleted.

2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm. Millimeters are in parentheses.

3/ Material shall be silver (MIL-S-13282 grade A) laminated on copper alloy, minimum thickness .005 inch(.13mm).

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TABLE II. Engineering information.

Part number M85/3-	Suggested frequency range TE_{10} mode	Cutoff for TE_{10} mode	Theoretical 1/ attenuation (lowest to highest frequency)	Theoretical peak power rating 2/ (lowest to highest frequency)	Theoretical CW power rating 3/ (lowest to highest frequency)
	GHz	GHz	dB/100 ft	kilowatts	watts
006-XXX	26.50 - 40.00	21.10	24.55 - 16.80	96.0 - 146	103.1 - 150.1
007-XXX			23.02 - 15.77		109.7 - 160.1
008-XXX			21.99 - 15.06		115.1 - 168.0
009-XXX			34.46 - 23.59		73.27 - 107.0
010-XXX	33.00 - 50.00	26.350	34.57 - 23.50	64.4 - 97.0	64.73 - 95.30
011-XXX			32.44 - 22.05		68.89 - 101.4
012-XXX			30.98 - 21.06		72.29 - 106.3
013-XXX			48.53 - 32.99		46.05 - 67.74
014-XXX	40.00 - 60.00	30.69	42.39 - 30.46	48 - 70	48.30 - 67.21
015-XXX			39.81 - 28.60		51.32 - 71.43
016-XXX			38.02 - 27.32		53.85 - 74.94
017-XXX	50.00 - 75.00	39.90	64.23 - 43.89	30 - 40	28.46 - 41.44
018-XXX			60.25 - 41.17		30.27 - 44.30
019-XXX			57.55 - 39.32		31.76 - 46.49
020-XXX	60.00 - 90.00	48.40	87.79 - 58.86	20 - 30	19.15 - 28.56
021-XXX			82.37 - 55.22		20.37 30.38
022-XXX			78.67 - 52.74		21.37 - 31.88
023-XXX	75.00 - 110.00	58.85	112.5 - 79.26	14 - 20	13.82 - 19.63
024-XXX			105.6 - 74.37		14.73 - 20.86
025-XXX			100.9 - 71.03		15.40 - 21.88
026-XXX	90.00 - 140.00	73.84	171.4 - 110.0	8.8 - 13	8.618 - 13.43
027-XXX			160.9 - 103.3		9.161 - 14.27
028-XXX			153.7 - 98.67		9.610 - 14.97
029-XXX	110.00 - 170.00	90.845	238.1 - 150.9	5.9 - 9.3	5.662 - 8.934
030-XXX			223.5 - 141.7		6.019 - 9.494
031-XXX			213.5 - 135.3		6.315 - 9.964
032-XXX	140.00 - 220.00	115.750	343.6 - 216.0	3.7 - 6.1	3.674 - 5.844
033-XXX			322.7 - 202.8		3.904 - 6.211
034-XXX			308.2 - 193.7		4.096 - 6.517
035-XXX	170.00 - 260.00	137.52	428.2 - 283.2	2.8 - 4.5	2.832 - 4.282
036-XXX			402.0 - 265.9		3.010 - 4.551
037-XXX			384.0 - 254.0		3.484 - 4.774
038-XXX	220.00 - 325.00	173.28	570.9 - 388.0	1.9 - 2.6	2.021 - 2.973
039-XXX			536.0 - 364.4		2.148 - 3.159
040-XXX			512.0 - 348.0		2.253 - 3.315
041-XXX	90.0 - 140.0	73.84	171.4 - 110.0	8.8 - 13	6.442 - 10.04
042-XXX			160.9 - 103.3		6.847 - 10.67
043-XXX			153.7 - 98.67		7.183 - 11.19
044-XXX	110.0 - 170.0	90.845	238.1 - 150.9	5.9 - 9.3	4.238 - 6.688
045-XXX			223.5 - 141.7		4.506 - 7.107
046-XXX			213.5 - 135.3		4.727 - 7.459
047-XXX	140.0 - 220.0	115.750	343.6 - 216.0	3.7 - 6.1	2.666 - 4.241
048-XXX			322.7 - 202.8		2.833 - 4.507
049-XXX			308.2 - 193.7		2.972 - 4.729

See footnotes at end of table.

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TABLE II. Engineering information. - Continued

Part number M85/3-	Suggested frequency range TE_{10} mode	Cutoff for TE_{10} mode	Theoretical 1/ attenuation (lowest to highest frequency)	Theoretical peak power rating 2/ (lowest to highest frequency)	Theoretical CW power rating 3/ (lowest to highest frequency)
	GHz	GHz	dB/100 ft	kilowatts	watts
050-XXX	170.0 - 260.0	137.52	428.2 - 283.2	2.8 - 4.5	2.012 - 3.042
051-XXX			402.0 - 265.9		2.139 - 3.233
052-XXX			384.0 - 254.0		2.475 - 3.392
053-XXX	220.0 - 325.0	173.28	570.9 - 388.0	1.9 - 2.6	1.395 - 2.053
054-XXX			536.0 - 364.4		1.483 - 2.181
055-XXX			512.0 - 348.0		1.556 - 2.289

1/ At 20°C waveguide temperature.

2/ These values were determined by calculating the unpressurized air dielectric breakdown strength produced by considering the E field within the waveguide. This determination was based on the peak value of a continuous wave (CW) signal. For further information or to calculate for other situations, see Gould and Gilden's "Handbook of High Power Capabilities of Waveguide Systems" (available from Microwave Associates, Burlington, Massachusetts).

3/ These values were determined by calculation of the rate of heat loss to the ambient, considering a nonpressurized air dielectric waveguide in air using no artificial heat sink. A maximum waveguide temperature of 71°C, 1:1 VSWR, and an ambient temperature of 30°C were assumed. For further information or to calculate for other situations, see H. E. King, "Rectanuglar Waveguide Theoretical CW Average Power Rating", IRE Transactions PGMTT-9 pp 349-357, July 1961.

TABLE III. Supersession data and cross index.

Part number M85/3-	Superseded part number M85/1	Cross index		
		AN type	NATO type	EIA type
006-XXX	114-XXX	RG-96/U	NWG-6A22	WR28S
008-XXX	117-XXX	RG-271/U	---	---
010-XXX	118-XXX	RG-97/U	NWG-6A23	WR22S
012-XXX	121-XXX	RG-272/U	---	---
016-XXX	124-XXX	RG-358/U	---	---
017-XXX	125-XXX	RG-98/U	NWG-6A25	WR1SS
019-XXX	128-XXX	RG-273/U	---	---
020-XXX	129-XXX	RG-99/U	NWG-6A26	WR12S
022-XXX	132-XXX	RG-274/U	---	---
025-XXX	135-XXX	RG-359/U	---	---
043-XXX	138-XXX	RG-278/U	---	---
046-XXX	141-XXX	RG-276/U	---	---
049-XXX	144-XXX	RG-275/U	---	---
052-XXX	147-XXX	RG-277/U	---	---

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TABLE IV. Typical resistivity and conductivity of materials.

Material	Resistivity @ 20°C (max)	Conductivity
	microhm-cm	percent
Aluminum alloy 6061	4.00	40 to 45
Copper:		
OF	1.77	101
DLP	1.793	up to 100
Copper alloy	3.9	44
Silver:		
Grade A	1.63	106
Grade C	2.03	85

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI, SL, ME
Navy - OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC, AT
Navy - MC, SH, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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